

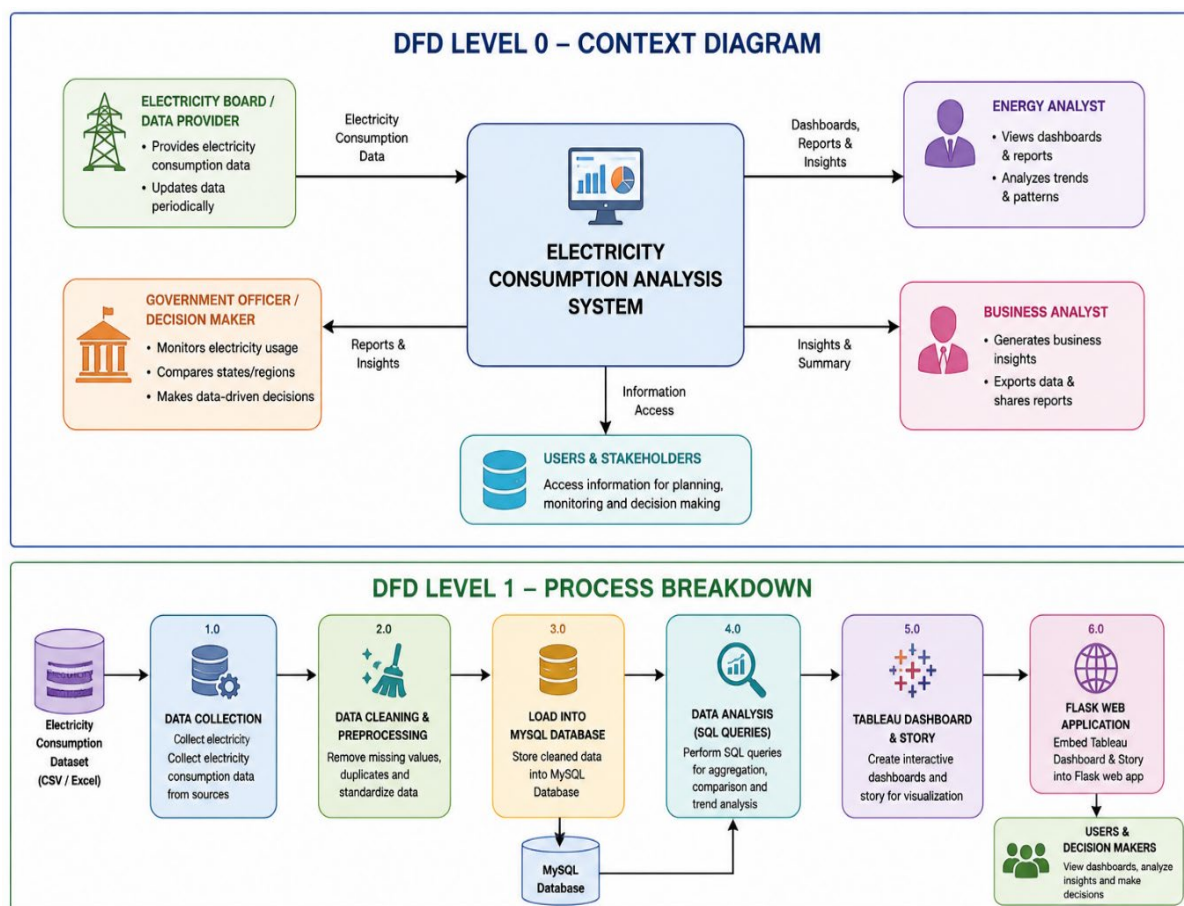
REQUIREMENT ANALYSIS

Data Flow Diagram & User Stories

Date	01 JULY 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Data Flow Diagrams

A **Data Flow Diagram (DFD)** is a graphical representation of how data moves through a system. It illustrates how data enters the system, how it is processed, where it is stored, and how the processed information is presented to users. For this project, the DFD describes the flow of electricity consumption data from the dataset to the MySQL database, Tableau dashboards, and the Flask web application.



USER STORIES

User Type	Functional Requirement (Epic)	User Story No	User Story / Task	Acceptance Criteria	Priority	Release
Energy Analyst	Dashboard KPIs	USN-1	As an Energy Analyst, I want to view total electricity usage and summary KPIs so that I can quickly understand electricity consumption.	KPI cards display accurate electricity statistics.	High	Sprint-1
Energy Analyst	State-wise Analysis	USN-2	As an Energy Analyst, I want to compare electricity usage across states so that I can identify high electricity consuming states.	State-wise chart displays correct values.	High	Sprint-1
Energy Analyst	Region-wise Analysis	USN-3	As an Energy Analyst, I want to compare electricity usage by region so that I can understand regional demand.	Region chart updates correctly.	High	Sprint-1
Energy Analyst	Monthly Trend	USN-4	As an Energy Analyst, I want to analyze monthly electricity consumption trends so that I can monitor seasonal changes.	Monthly trend graph displays correctly.	High	Sprint-2
Energy Analyst	Interactive Filters	USN-5	As an Energy Analyst, I want to filter data by state and region so that I can perform customized analysis.	All charts update when filters are selected.	High	Sprint-2
Government Officer	State-wise Report	USN-6	As a Government Officer, I want to identify high electricity consuming states so that I can plan electricity distribution effectively.	Dashboard shows accurate state ranking.	High	Sprint-2
Government Officer	Region-wise Report	USN-7	As a Government Officer, I want to compare electricity demand across regions for better planning.	Region comparison works correctly.	High	Sprint-2

User Type	Functional Requirement (Epic)	User Story No	User Story / Task	Acceptance Criteria	Priority	Release
Government Officer	Usage Map	USN-8	As a Government Officer, I want to visualize electricity consumption on a map so that I can easily understand geographic distribution.	Map displays all states correctly.	Medium	Sprint-3
Government Officer	Monthly Reports	USN-9	As a Government Officer, I want to monitor monthly electricity consumption so that I can identify demand changes.	Monthly chart displays correct trends.	Medium	Sprint-3
Business Analyst	Business Insights	USN-10	As a Business Analyst, I want to analyze electricity usage patterns so that I can generate useful business insights.	Dashboard displays meaningful insights.	High	Sprint-2
Business Analyst	Tableau Story	USN-11	As a Business Analyst, I want to present insights through a Tableau Story so that stakeholders can easily understand the analysis.	Story displays correctly with all story points.	Medium	Sprint-3

Conclusion

The Data Flow Diagram illustrates how electricity consumption data moves through the system from data collection to visualization and reporting. The User Stories capture the functional requirements of different users, including Energy Analysts, Government Officers, Business Analysts, and Dashboard Users. Together, they ensure that the Electricity Consumption Analysis system is user-centered, efficient, and capable of supporting data-driven decision-making through interactive dashboards and visual analytics.